Project Initialization and Planning Phase

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Problem Statement — Farmer
I am a small/medium-scale farmer trying to decide which crop to plant this season, but I have no fast, reliable way to match my soil's actual nutrient levels and the local climate to the right crop, because crop selection is typically based on tradition or informal advice rather than measured data, which makes me feel anxious about financial loss if I choose poorly.

Problem Statement — Agricultural Researcher / Policymaker
I am an agricultural researcher trying to study crop-environment relationships across many samples, but I lack an accessible tool that lets me explore how soil and climate parameters relate to crop suitability at scale, because most existing tools are either single-prediction demos or require manual statistical analysis, which makes me feel limited in producing timely, evidence-based recommendations for sustainable farming policy.

Problem Statement — General
Modern agriculture suffers from poor crop selection, inefficient resource utilization, and reduced productivity, largely because farmers lack accessible, data-driven decision support tools that translate raw soil and climate measurements into a clear, actionable crop recommendation.